Development of Design Procedures for the Synthesis of Design Solutions for Data Management, Design and Production Procedures at the Stages of the Life Cycle of an Electronic Product

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***Abstract*— The report suggests the development of design procedures for the synthesis of design solutions for managing product data and design and production procedures at the stages of the life cycle. Variants of the tasks of forming design decisions are given, the requirements for them are formulated. The rules of choosing descriptions of the components of a digital passport and the wording of the similarity criterion are presented, which forms the developed design procedures. Based on the results obtained, an implementation option is proposed for the formation of design decisions.**

Keywords*— *design procedures; data management; synthesis of design decisions; management of design and production procedures

I. Introduction

A modern electronic product is a complex hardware-software complex, the delivery of which is based on the application of the customer. It is analyzed, identifying lists of work, deadlines and performers, both from among the departments of the enterprise and organizations, which sets the task of operational interaction between them. This necessitates the exchange of relevant product data, for which PDM, ERP, MES and / or QMS systems are used [1-4]. Their diversity allows to choose those solutions that satisfy the requirements of enterprises. As a result, each company creates its own unique description of the product in the form of its digital passport - a set of product data - information objects of the systems - and the corresponding design and production procedures [5,6].

The content of the digital passport allows to implement the interaction between departments of one enterprise, but it is not as effective for solving the problem of interaction between enterprises. Common methods are to implement the same systems with synchronized versions and models of stored data for all participants in the interaction [7-11]. Another way is to transfer data using a neutral STEP data exchange format. However, they both have one obvious drawback - the organization of electronic interaction between enterprises is possible only with the coordination of the formats of the transmitted data.

Therefore, the task of developing new approaches to the organization of interaction between various enterprises, in essence, comes down to the interaction of heterogeneous systems and is relevant. As an approach to its solution, it is proposed to use the contents of a digital passport to manage product data and design and production procedures. For this, such works are implemented as: analysis of the tasks of forming design decisions; generalization of the requirements for them; development of design procedures for the synthesis of design solutions at the stages of the product life cycle.

This work focuses on the results of creating procedures for the synthesis of design solutions for data management, design and production procedures, which includes the development of rules for choosing descriptions of the components of a digital passport and the formation of a similarity criterion for design decisions.

II. TASK VARIANTS FOR PROJECT SOLUTIONS FORMATION

An analysis of the managing product data processes about the product, design and production procedures at the stages of the life cycle showed that the digital passport consists of k sets of components,

$$C_1 = (c_{11}, c_{12}, c_{13}, ..., c_{1n_1}),$$
$$C_2 = (c_{21}, c_{22}, c_{23}, ..., c_{2n_2}),$$
$$\dots$$
$$C_k = (c_{k1}, c_{k2}, c_{k3}, ..., c_{kn_k}),$$

the elements of which are given in pairs "information object - design and production procedure", such that

$$c_{ij} = < D_i, \Pr_j >, \; i = 1...n, \; j = 1...n_k,$$

where D_i – data type on the electronic product or type of information object;

and Pr_j – design and production procedure.

In addition, the content of the passport establishes a link between the stage of the life cycle and the corresponding design and production procedure. Their implementation requires the adoption of certain design decisions, for which the following final set of tasks has been formed and generalized:

1) search for data on products and their components;
2) search for data on purchased components (PC), materials, products of interplant cooperation (IPC) and their suppliers;
3) data generation for developed or revised documentation;
4) search for data on design, software and technological documentation (DD, SD and TD) on the components of the product to be developed or adjusted;
5) monitoring the implementation of the positions of unified tasks (operational work schedules and applications for launching production) [12];
6) management of production operations;
7) search for design and production procedures.

Studies have shown that the resulting list of tasks allows to vary the generated design decisions needed to manage product data, design and production procedures. Moreover, the variability is determined by the requirements, also divided into two types.

III. REQUIREMENTS FOR DESIGN SOLUTIONS

Requirements contain indications of parameters whose values must be analyzed to form a design solution that corresponds to the task in the design and production procedure at the stage of the life cycle.

Then, the first type - the requirements for managing product data - include:

1) product type - a complex, assembly unit, part, kit;
2) the manufacturer of the product - purchased, own production;
3) product feature - IPC, newly developed, borrowed;
4) presence of comments on the DD - no comments, there are unworked comments, there are worked-out comments;
5) presence of comments on the SD - there are no comments, there are unworked comments, there are worked-out comments;
6) presence of comments on the TD - there are no comments, there are unworked comments, there are worked-out comments;
7) the possibility of acquisition - there are several suppliers, there is a single supplier;
8) a sign of products of inadequate quality - there are no quality problems, there are small quality problems, there are serious quality problems;
9) a sign of availability in stock - not in stock, in stock;
10) document type - list of elements, specification sheet, specification, spare parts list, electronic documents sheet, electronic product structure (EPS), list of purchased products, software specification;
11) type of EPS - functional EPS, structural EPS, industrial and technological EPS, physical EPS, operational EPS.

The second type - the requirements for design procedures - include::

1) type of document - operational schedule of work, application for the start of production;

2) type of work - development of DD and SD, development of TD, adjustment of DD and TD, purchase, production;

3) controlled condition - “Under development”, “At verification”, “Delivered to the archive”, “Completed”;

4) type of product - a complex, assembly unit, part, kit;

5) the presence of comments on the DD - no comments, there are unworked comments, there are worked-out comments;

6) the presence of comments on the TD - there are no comments, there are unworked comments, there are worked-out comments;

7) availability in stock feature - not in stock, in stock;

8) production feature - started, suspended, completed.

The latter type - the requirements for production procedures - include::

1) stage of the product life cycle - conclusion of an agreement, development, preparation of production, production, operation, repair;

2) type of document from the customer - application for development, application for supply, reclamation certificate, application for service;

3) the task of forming a design solution - a list of tasks listed previously.

Then, on the basis of the foregoing, it is logical to conclude that the requirements indicate the area within which the desired design solutions should be located. Moreover, the requirements are set in the form of linguistic variables [13] with one of the values - “0” (if the parameter value is not analyzed) or “1” (otherwise), and decisions are formed from the contents of the digital passport.

Thus, it is possible to develop analysis procedures and rules for selecting components of a digital passport.

IV. PROCEDURE FOR ANALYSIS OF DESIGN DECISIONS

Studies have shown that the components of a digital passport are interconnected and can be represented by a mathematical model [14] of the form:

a component model $R(C)$, dimension $k \times n_{k}$, which presents the correspondence between the types of data on the electronic product (information objects) and design and production procedures, which forms the pair "object - design and production procedure";

a description model $G(C)$, dimension $k \times P$, where each stage of the product's life cycle is associated with a pair of "information object - design and production procedure";

substitution model $Z(C)$, dimension $k \times k$ each element of which is equal to "1" if two components of the digital passport correspond to each other or "0" otherwise;

interaction model $X(C)$, dimension $k \times k$, each element of which is equal to "1" if two components of a digital passport are connected to each other or "0" otherwise;

a model of the parameters of the components of the digital passport U, corresponding to the lists presented earlier for the requirements.

In this case, the task of creating design decisions is reduced to constructing a matrix $X(C)$, the elements of which are calculated on the basis of models of descriptions and parameters given, as mentioned above, by linguistic variables.

Then, the procedure for analyzing design decisions consists of a sequence of the following actions:

1. Create a list of parameters of the components of the digital passport $\vec{U}$ and the components $\vec{C}$ themselves, used at a certain stage of the life cycle to manage data or design and production procedures.
2. Construct matrices of pairwise comparisons of elements of the set $\vec{U}$ for each element of the set C in accordance with the method proposed by Chernov V. [13].
3. It is proved that the values of the membership function for a linguistic variable can be calculated through the eigenvector of a matrix of the form

$$N \cdot \vec{\omega} = \lambda \cdot \vec{\omega}$$

where $\vec{\omega} = (\vec{\omega_1}, \vec{\omega_2}, \vec{\omega_3}, ..., \vec{\omega_P})$ – matrix eigenvector, which is a membership function value; λ – matrix eigenvalue.

4. Actions 2 and 3 are performed until the identity of the parameters for each component of the digital passport is determined.

V. RULES FOR SELECTING DESCRIPTIONS OF DIGITAL PASSPORT COMPONENTS

The results of the analysis of design decisions allow us to develop rules for choosing descriptions of the components of a digital passport in the form of the following sequence of steps [15]:

1. Create a set of rules for checking the proximity of descriptions of the components of the digital passport $G(\vec{C})$, the requirements of PR. The list of rules contains as many of them as there are elements in the set $\vec{C}$.
2. Check the implementation of each rule. Verification means solving a linear equation of the form

$$G_i(\overrightarrow{C_i}) = a + \sum_{j=1}^{P} b_j \cdot RP_j \cdot g_{ij}(\overrightarrow{C_i}); i = \overline{1,K}; K \le k .$$

3. Make a selection of descriptions of the components of the digital passport that satisfy the given conditions, in accordance with the expression

$$G^*(\vec{C}) = \beta \cdot X + \varepsilon \text{ ,}$$

where X – deterministic matrix of dimension $K \times P$, such that $X = RP \cdot G$; β – column vector containing elements $\beta = (a, b_1, b_2, ..., b_K)$.

4. Verify the fulfillment of the Gauss-Markov conditions to confirm the possibility of forming design decisions in accordance with the similarity criterion specified by the least squares method.

Thus, the procedure for selecting descriptions of the components of a digital passport is reduced to searching for the elements of the column vector β.

VI. CRITERIA FOR SIMILARITY OF DESIGN DECISIONS

Thus, the fulfillment of the Gauss-Markov conditions allows us to formulate a similarity criterion of the following form.

$$\Delta(\beta) = \sum_{i=1}^{K} (G_i(\vec{C}) - G_i^*(\vec{C}))^2 =$$
$$= (G - \beta \cdot X)^T \cdot (G - \beta \cdot X) \to \min$$

As a result of transformations of the expression obtained

$$\frac{\partial \Delta(\beta)}{\partial \beta} = 2 \cdot \beta \cdot X^T \cdot X - 2 \cdot X^T \cdot G = 0$$
$$\beta = (X^T \cdot X)^{-1} \cdot (X^T \cdot G) \text{ .}$$

In addition, the upper and lower boundaries [16] of the value range are determined G_i^*, $i = \overline{1,K}$, such that

$$G^*_{max} < G^*_i < G^*_{min},$$
$$G^*_i - S < G^*_i < G^*_i + S,$$

where

$$S = S_{ocm} \cdot \sqrt{1 + \frac{1}{K} + \frac{(X_i - \overline{X})^2}{\sum_{i=1}^{K}(X_i - \overline{X})^2}},$$

$$S_{ocm} = \sqrt{\frac{\sum_{i=1}^{K}(G_i - G^*_i)^2}{K-2}},$$

$$\overline{X} = \frac{\sum_{i=1}^{K} X_i}{K}.$$

The results obtained allow us to solve the problem of forming design decisions in an automated way.

VII. Option for implementing the problem forming project solutions

Since the mathematical model of a digital passport contains an interaction model, it is logical to state:

1) the problem being solved is graph;

2) the components of the digital passport form a tree.

Therefore, the problem to be solved can be reduced to the construction of a spanning tree [17].

It is known that the best results of its solution are achieved using algorithms based on the analysis of flows in networks [17], genetic algorithms and methods of evolutionary modeling [18-21]. The use of one of them involves the formation of descriptions of the components of a digital passport according to the similarity criterion, as well as their subsequent filtering, taking into account the upper and lower boundaries of the value range.

Therefore, the problem is solved when, out of all possible options for the content of the digital passport, its components are determined that form the design decision in accordance with the similarity criterion. This requires the implementation of several stages of the algorithm based on the bee swarm method [21]:

1. The formation of the search space for a design solution in accordance with the similarity criterion.
2. Search for promising positions that fall within the confidence interval specified by the upper and lower bounds of values.
3. The choice of base positions that satisfy the condition

$$\Delta(\beta) \to \min.$$

4. Formation of a new set by removing an arbitrary number of elements, for which the values of the column vector β are calculated, as well as the conversion of the values of G^* and $\Delta(\beta)$ for the elements of the set obtained in the previous step.
5. Formation of a new search space in accordance with the similarity criterion.

The algorithm ends if the spanning tree protocol is completed, or if all the vertices are analyzed - the components of a digital passport.

VIII. Conclusion

Thus, the design procedures for the synthesis of design decisions have been developed, containing the rules for choosing the components of a digital passport and a similarity criterion for design decisions. Variants of tasks of forming design decisions and lists of requirements for them are presented. The results obtained made it possible to develop an algorithm based on the paradigm of behavior of a bee colony. And this, in its turn, is a methodology for using a digital passport to formulate design solutions in the instrumentation manufacturing industry.

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